

# Esercizi Lez. 13 - Teoria

**Esercizio 1** - Calcolare la FGM delle variabili aleatorie:

Discrete:

$X \in \{1, 2, 3, 4, 5, 6\}$  Uniforme Discreta

$$G_X(t) = E[e^{tx}] = \sum_{a \in \{1, \dots, 6\}} e^{ta} \cdot f(a) = \sum_{a \in \{1, \dots, 6\}} e^{ta} \cdot \frac{1}{6} = \frac{1}{6} \cdot \sum_{a \in \{1, \dots, 6\}} e^{ta} = \boxed{\frac{1}{6} \cdot \frac{e^t(1-e^{6t})}{1-e^t}}$$

$Y \sim \text{Ber}(p)$

$$G_Y(t) = E[e^{tx}] = \sum_{a \in \{0, 1\}} e^{ta} \cdot p_Y(a) = e^{t \cdot 0} \cdot (1-p) + e^{t \cdot 1} \cdot p = \boxed{e^t p + (1-p)}$$

$Z \sim \text{Geom}(p)$

$$\begin{aligned} G_Z(t) &= E[e^{tx}] = \sum_{K=1}^{\infty} e^{tK} \cdot p_X(K) = p \cdot \sum_{K=1}^{\infty} e^{tK} \cdot (1-p)^{K-1} \\ &= p \cdot \sum_{K=1}^{\infty} (e^t)^{K-1+1} \cdot (1-p)^{K-1} = p \cdot e^t \cdot \sum_{K=1}^{\infty} (e^t \cdot (1-p))^{K-1} \\ &= p \cdot e^t \cdot \sum_{M=0}^{\infty} (e^t \cdot (1-p))^M \quad \begin{matrix} e^t(1-p) < 1 \\ \Rightarrow \end{matrix} p \cdot e^t \cdot \frac{1}{1-e^t(1-p)} = \begin{cases} \frac{p \cdot e^t}{1-e^t(1-p)} & \text{se } t < \ln\left(\frac{1}{1-p}\right) \\ \infty & \text{se } t \geq \ln\left(\frac{1}{1-p}\right) \end{cases} \end{aligned}$$

Continue:

$Z \sim N(0,1)$

$\mu=0, \sigma^2=1$

$$\begin{aligned} G_Z(t) &= E[e^{tx}] = \int_{-\infty}^{+\infty} e^{tx} \cdot f(x) \cdot dx = \int_{-\infty}^{+\infty} e^{tx} \cdot \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}} \cdot dx = \frac{1}{\sqrt{2\pi}} \cdot \int_{-\infty}^{+\infty} e^{tx} \cdot e^{-\frac{x^2}{2}} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}} \cdot \int_{-\infty}^{+\infty} e^{tx - \frac{x^2}{2}} \cdot dx = \frac{1}{\sqrt{2\pi}} \cdot \int_{-\infty}^{+\infty} e^{-\frac{1}{2}(x^2 - 2tx)} \cdot dx = \frac{1}{\sqrt{2\pi}} \cdot \int_{-\infty}^{+\infty} e^{-\frac{1}{2}[(x-t)^2 - t^2]} \cdot dx \\ &= \frac{1}{\sqrt{2\pi}} \cdot \int_{-\infty}^{+\infty} e^{-\frac{(x-t)^2}{2}} \cdot e^{\frac{t^2}{2}} \cdot dx = e^{\frac{t^2}{2}} \cdot \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{(x-t)^2}{2}} \cdot dx = \boxed{e^{\frac{t^2}{2}}} \end{aligned}$$

$$x^2 - tx = x^2 - tx + t^2 - t^2 = (x-t)^2 - t^2$$

$$\begin{matrix} N(0,1) \text{ con } y=x-t \\ \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{y^2}{2}} \cdot dy = 1 \end{matrix}$$

## $X \sim \text{Pois}(u)$

$$G_X(t) = E[e^{tx}] = \sum_{m=0}^{\infty} e^{tm} \cdot p_X(m) = \sum_{m=0}^{\infty} e^{tm} \cdot \frac{\mu^m \cdot e^{-\mu}}{m!} = e^{-\mu} \cdot \sum_{m=0}^{\infty} \frac{(\mu \cdot e^t)^m}{m!}$$

SERIE DI TAYLOR

$$= e^{-\mu} \cdot e^{\mu e^t} = e^{\mu(e^t - 1)} \quad \forall t \in \mathbb{R}$$

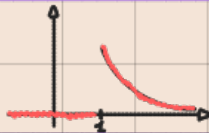
POISSON

$$p_X(m) = \frac{\mu^m}{m!} \cdot e^{-\mu}$$

## $X \sim \text{Par}(\alpha)$ , per quali valori di $t$ , $G(t)$ è finita?

PARETO

$$f(x) = \frac{\alpha}{x^{\alpha+1}} \cdot \mathbb{1}_{(1, +\infty)}(x) = \begin{cases} \frac{\alpha}{x^{\alpha+1}} & \text{SE } x \in (1, +\infty) \\ 0 & \text{SE } x \notin (1, +\infty) \end{cases}$$



$$G_X(t) = E[e^{tx}] = \int_1^{+\infty} e^{tx} \cdot \frac{\alpha}{x^{\alpha+1}} \cdot dx$$

SE  $t \geq 0$ :

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^{\alpha+1}} = +\infty \Rightarrow \text{ALLORA L'INTEGRALE SICURAMENTE DIVERGE} \Rightarrow \alpha \cdot \int_1^{+\infty} \frac{e^{tx}}{x^{\alpha+1}} \cdot dx = +\infty \Rightarrow G_X(t) = +\infty$$

SE  $t < 0$ :

$$\alpha \cdot \int_1^{+\infty} \frac{e^{tx}}{x^{\alpha+1}} \cdot dx < +\infty \Rightarrow G_X(t) < +\infty$$

ABBIAMO DIMOSTRATO CHE:  $X \sim \text{Par}(\alpha) \Rightarrow G_X(t) < +\infty \iff t < 0$

## $X \sim \text{Exp}(\lambda)$

ESPOENZIALE

$$f(x) = \lambda e^{-\lambda x} \cdot \mathbb{1}_{(0, +\infty)}(x) = \begin{cases} \lambda e^{-\lambda x} & \text{SE } x \in (0, +\infty) \\ 0 & \text{SE } x \notin (0, +\infty) \end{cases}$$

$$G_X(t) = E[e^{tx}] = \int_0^{+\infty} e^{tx} \cdot \lambda e^{-\lambda x} \cdot dx = \lambda \cdot \int_0^{+\infty} e^{-(\lambda-t)x} \cdot dx$$

CASO 1)  $\lambda - t > 0$ :  $\lambda \cdot \int_0^{+\infty} e^{-(\lambda-t)x} \cdot dx = \frac{\lambda}{(\lambda-t)} \cdot \int_0^{+\infty} (\lambda-t) \cdot e^{-(\lambda-t)x} \cdot dx = \frac{\lambda}{\lambda-t} \cdot \overset{1}{\text{Exp}(\lambda-t)} = \frac{\lambda}{\lambda-t}$

CASO 2)  $\lambda - t \leq 0$ :  $\lambda \cdot \int_0^{+\infty} e^{-(\lambda-t)x} \cdot dx = +\infty$

QUINDI:

$$G_X(t) = \begin{cases} \frac{\lambda}{\lambda-t} & \text{SE } \lambda > t \\ +\infty & \text{SE } \lambda \leq t \end{cases}$$

## $X \sim N(\mu, \sigma^2)$

$$G_X(t) = E[e^{tx}] = \int_{-\infty}^{+\infty} e^{tx} \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \cdot dx = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{tx - \frac{(x-\mu)^2}{2\sigma^2}} \cdot dx$$

COMPLETIAMO IL QUADRATO NELL'ESPONENTE:  $tx - \frac{(x-\mu)^2}{2\sigma^2} = \frac{(x-\mu - \sigma^2 t)^2}{2\sigma^2} - \mu t + \frac{\sigma^2 t^2}{2}$

$$G_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}} \cdot \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{-\frac{(x-\mu - \sigma^2 t)^2}{2\sigma^2}} \cdot dx = e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

$$N(\mu + \sigma^2 t, \sigma^2) = 1$$

GAUSSIANA

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

## Esercizio 2 - Calcolare i momenti richiesti.

Calcolare il momento terzo di  $Z \sim N(0,1)$

$$E[X^3] = G_x'''(0)$$

$$G_x(t) = e^{\frac{t^2}{2}}$$

$$G_x'(t) = \left(e^{\frac{t^2}{2}}\right)' = e^{\frac{t^2}{2}} \cdot \frac{2t}{2} = t \cdot e^{\frac{t^2}{2}}$$

$$G_x''(t) = \left(t \cdot e^{\frac{t^2}{2}}\right)' = 1 \cdot e^{\frac{t^2}{2}} + t \cdot t e^{\frac{t^2}{2}}$$

$$G_x'''(t) = \left(e^{\frac{t^2}{2}} + t^2 \cdot e^{\frac{t^2}{2}}\right)' = t \cdot e^{\frac{t^2}{2}} + 2t \cdot e^{\frac{t^2}{2}} + t^3 \cdot e^{\frac{t^2}{2}} \\ = 3t \cdot e^{\frac{t^2}{2}} + t^3 \cdot e^{\frac{t^2}{2}}$$

$$G_x'''(0) = 3 \cdot 0 \cdot e^{\frac{0^2}{2}} + 0^3 \cdot e^{\frac{0^2}{2}} = 0$$

Calcolare il momento terzo di  $Y \sim \text{Exp}(\lambda)$

$$E[X^3] = G_x'''(0) \quad t < \lambda \quad \lambda > 0$$

$$G_x(t) = \begin{cases} \frac{\lambda}{\lambda - t} & t < \lambda \\ +\infty & t \geq \lambda \end{cases}$$

$$G_x'(t) = \left(\frac{\lambda}{\lambda - t}\right)' = \lambda \cdot \left(\frac{1}{\lambda - t}\right)' = \lambda \cdot \left(-\frac{(\lambda - t)'}{(\lambda - t)^2}\right) = \lambda \cdot \left(-\frac{-1}{(\lambda - t)^2}\right) = \frac{\lambda}{(\lambda - t)^2}$$

FATTORE COSTANTE

$$(c \cdot g(x))' = c \cdot (g(x))'$$

$$G_x''(t) = \left(\frac{\lambda}{(\lambda - t)^2}\right)' = \lambda \cdot \left(-2 \cdot \frac{1}{(\lambda - t)^3} \cdot (-1)\right) = \frac{2\lambda}{(\lambda - t)^3}$$

$$G_x'''(t) = \left(\frac{2\lambda}{(\lambda - t)^3}\right)' = 2\lambda \cdot \left(-3 \cdot \frac{1}{(\lambda - t)^4} \cdot (-1)\right) = \frac{6\lambda}{(\lambda - t)^4}$$

REGOLA DELLA FUNZIONE RECIPROCA

$$O\left[\frac{1}{v}\right] = O[v^{-1}] = O[-1 \cdot v^{-2}] = O\left[-\frac{1}{v^2}\right]$$

$$G_x'''(0) = \frac{6\lambda}{(\lambda - 0)^4} = \frac{6}{\lambda^3} \quad \text{CON } \lambda > 0$$

Calcolare il valore atteso di  $X \sim U(0,1)$  col metodo della FGM

$$G_x(t) = E[e^{tx}] = \int_0^1 e^{tx} \cdot f(x) \cdot dx = \int_0^1 e^{tx} \cdot \frac{1}{1-0} \cdot dx = \int_0^1 e^{tx} \cdot dx = \left.\frac{e^{tx}}{t}\right|_0^1 = \frac{e^t - 1}{t} \quad t \neq 0$$

$$G_x'(t) = \frac{t \cdot e^t - (e^t - 1)}{t^2} = \frac{e^t(t-1) + 1}{t^2}, \quad \lim_{t \rightarrow 0} G_x'(t) = \frac{(1 + \frac{t^2}{2} + \dots)(t-1) + 1}{t^2}$$

$t \rightarrow 0$   
NUMERATORE:  $(1 + t + \frac{t^2}{2})(t-1) + 1 = (t-1 + t^2 - t + \frac{t^3}{2} + \frac{t^2}{2}) + 1 = \frac{t^2}{2} + o(t^2)$

$$G_x'(t) \approx \frac{\frac{t^2}{2}}{t^2} = \frac{1}{2}$$

MOMENTI ESPONENZIALE

$$E[X^m \sim \text{Exp}(\lambda)] = \frac{m! \cdot \lambda}{(\lambda - t)^{m+1}} \xrightarrow{t=0} \frac{m!}{\lambda^m}$$

MOMENTI NATURALE STANDARD

$$E[Z^{2K} \sim N(0,1)] = (2K-1)!! = \frac{(2K)!}{2^K \cdot K!}$$

$$E[Z^{2K+1} \sim N(0,1)] = 0$$

La prof all'esame può chiedere di calcolare la FGM,  $G_x(t)$ , per una variabile aleatoria non nota! Quindi é importante esercitarsi anche su questi.